Airport Data Analysis Using SQL

Uncovering patterns in airport performance, airline operations, and route demand to drive strategic decisions

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Project Overview

Route Analysis

High-traffic corridors and passenger flow patterns

Airline Performance

Market share and operational efficiency

Passenger Traffic

Volume trends across the network

Generate actionable insights for route planning, capacity allocation, and infrastructure optimization

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Dataset & Business Questions

Dataset Contains

- Passenger counts and freight data
- Origin/destination airports
- Time attributes (year, quarter, month)
- Airline identifiers

Key Questions

- Top routes by passenger volume
- Airline market share analysis
- Underutilized routes identification
- Growth potential airports

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Top 10 Busiest Routes

1	**Miami → New York** 156,277 passengers
2	**New York → Miami** 149,209 passengers
3	**Orlando → New York** 130,145 passengers
4	**New York → Orlando** 125,937 passengers
5	**New York → Chicago** 121,251 passengers

Florida-NYC corridor dominates with 4 of top 5 routes

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Airline Market Share

11M

Southwest

Market leader

9.7M

Delta

Second place

9.5M

American

Third place

8.1M

United

Fourth place

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SQL Techniques Used

Common Table Expressions

CTEs for modular query structure

Window Functions

ROW_NUMBER, RANK for analysis

Aggregations

SUM, AVG, COUNT operations

Joins & UNION

Combining data sources

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Key Insights

Concentrated Demand

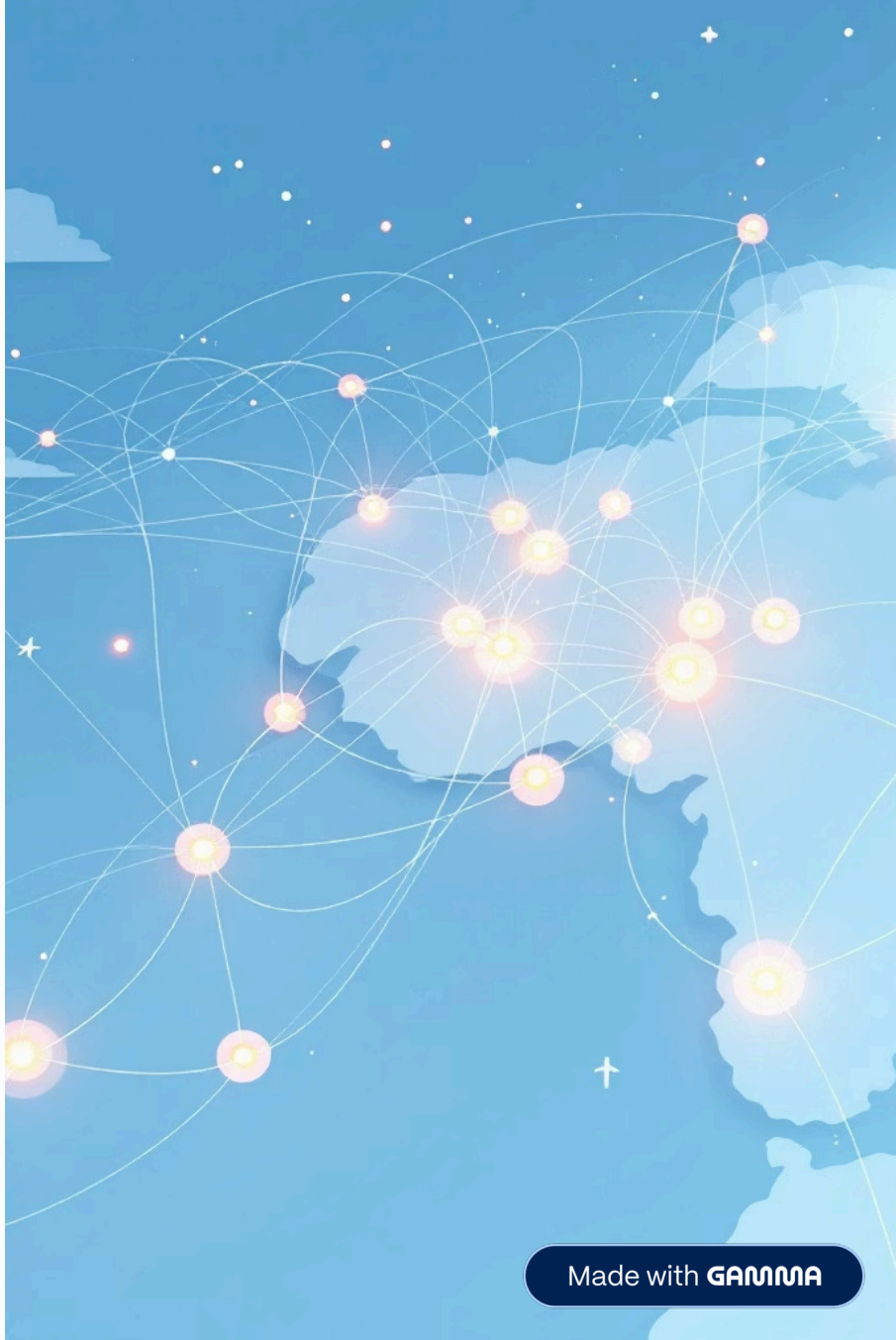
Specific city pairs show high passenger volumes

Hub Performance

Major airports handle disproportionate traffic

Growth Potential

Underutilized routes identified



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Underutilized Routes

High Frequency, Low Passengers

Alaska routes show very low average passengers per flight despite regular service

- Anchorage → Miami: 0 passengers
- Fairbanks → Rampart: 4.8 avg
- Multiple Bethel connections: 23-29 avg

Opportunity

Marketing campaigns and smaller aircraft could improve efficiency

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Business Recommendations

01

Increase Capacity

Deploy larger aircraft on high-demand routes

02

Strengthen Hubs

Invest in infrastructure at major airports

03

Improve Efficiency

Market underutilized routes, use smaller planes

04

Target Growth

Dynamic pricing and campaigns for potential areas

05

Optimize Operations

Top airlines should maintain service quality

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Next Steps

Implement Changes

Deploy capacity increases on top routes

Monitor Performance

Track passenger growth on targeted routes

Refine Strategy

Adjust based on real-time data

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